Overall, the statistical analysis suggests that the Rosiglitazone treatment is most effective at reducing the onset of diabetes in pre-diabetics. In summary, the clinical trial consisted of 699 patients, and these patients were (randomly?) assigned to three treatments: lifestyle (diet and exercise) adjustments (234 people), Metformin (232 people), and Rosiglitazone (233 people). Of those assigned lifestyle changes, 125 successfully avoided diabetes diagnosis, which gives a success rate of 53.4%. Alternatively, 112 successfully avoided diabetes diagnosis with Metformin (success rate of 48.3%), while 143 avoided diabetes for a success rate of 61.3%. The entire summary can be seen in Table 1 below.

To support these pointwise estimates, 10,000 sample bootstraps were constructed for each group, with KDEs and ECDFs being generated to visualize these results. As expected, each treatment group is centered roughly at its pointwise estimate, and they each appear symmetric about their mode; the majority of the density sits within 0.075 of the mode for each case. Still, there are some subtle differences. For one, the Metformin distribution has the fattest tails (particularly the lower tail). Since the range of the middle 90% of the data for all three distributions is roughly the same (0.108, 0.107, and 0.103), these fat tails appear to be driven mostly by a small set of outliers in the bootstrap. On the other hand, Rosiglitazone has the highest mode, suggesting that its density is mostly concentrated near the center of the distribution; the KDE also has a slanted peak, which could be an artifact of the original sample. For the full visual, see Figure 1 below.

The bootstrapped ECDFs convey a similar message but perhaps more explicitly. Indeed, Rosiglitazone demonstrates a first-order stochastic dominance of lifestyle and Metformin, and lifestyle demonstrates a first-order stochastic dominance of Metformin. In other words, the probability of obtaining a bootstrap sample with a certain success rate or higher is always as high or higher for the Rosiglitazone group than the lifestyle group or the Metformin group; similarly, the probability of obtaining a bootstrap sample with a certain success rate or higher is always as high or higher for the lifestyle group than the Metformin group. The two visuals also suggest the lifestyle group is "closer" in distribution to Metformin than Rosiglitazone. That is, for a given success rate, the lifestyle treatment's empirical probability of obtaining a sample with that success rate or lower is closer in magnitude to Metformin's probability than Rosiglitazone’s. More practically, these visuals suggest a hierarchical ordering of success rate, where Rosiglitazone tends to have the highest success rate, followed by lifestyle and Metformin; the discrepancy between lifestyle and Metformin is smaller than the discrepancy between Rosiglitazone and lifestyle, which communicates the message that there are possibly two “tiers” of treatment success, with Rosiglitazone being in a league of its own. For the full visual, see Figure 1 below.

Constructing ECDFs and KDEs for the pairwise treatment effects makes it easier to statistically justify the findings from the previous paragraph. For each pairwise treatment effect, three confidence intervals (90%, 95%, and 99%) were constructed. For a k % confidence interval, if we were to perform the same bootstrapping procedure many times, we would expect k% of the constructed intervals to contain the true average test effect. Since we only perform the

procedure once, we are not guaranteed that the true average test effect lies in our interval, nor are we told where in the interval it would be most likely to lie. Instead, we should consider the values within our interval as plausible.

For the average treatment effect between lifestyle and Metformin, each of the three confidence intervals displayed contains 0, which means that 0 is a plausible average treatment effect for all significance thresholds. Thus, we would fail to reject the null hypothesis at each threshold. Explicitly, there is no evidence to suggest that the average treatment effect of the lifestyle and metformin treatments is different. For the average treatment effect between lifestyle and Rosiglitazone, the 90% confidence interval does not contain 0 (it is entirely negative), whereas the 95% and 99% confidence intervals do contain 0; 0 is not a plausible treatment effect for the former interval but is a plausible effect for the latter two intervals. Thus, we would reject the null hypothesis at a significance level of 0.10, but reject at a significance level of 0.05 and 0.01. Explicitly, there is some evidence to suggest that the average treatment effect of the lifestyle and Rosiglitazone treatments is different, namely that the Rosiglitazone treatment is more effective. For the average treatment effect between Metformin and Rosiglitazone, each of the three confidence intervals displayed does not contain 0 (they are entirely negative), which means that 0 is not a plausible average treatment effect for all significance thresholds. Thus, we reject the null hypothesis at each threshold. Explicitly, there is strong evidence to suggest that the average treatment effects of the Metformin and Rosiglitazone treatments are different, namely that the Rosiglitazone treatment is more effective. For exact intervals, consider Table 2 below; for exact bootstrap distributions, consider Figures 2, 3, and 4 below.

In conclusion, the data generally suggest that Rosiglitazone is the superior intervention strategy for pre-diabetics, with one significance threshold suggesting that there is a meaningful difference in Rosiglitazone's success rate compared to lifestyle and three significance thresholds suggesting that there is a meaningful difference in Rosiglitazone's success rate compared to Metformin. While lifestyle changes had a better empirical success rate than Metformin, none of the significance thresholds deemed the difference meaningful. Thus, the data imply a two-tier hierarchy for diabetic intervention.

Even with this statistical analysis, the experimental design leaves some doubt. For one, the assignment instructions do not make it explicit whether subject randomization was applied. If individuals were able to choose their groups themselves, it is possible that the uncontrolled attributes of the two groups are not the same on average. For instance, people with a worse life prognosis may choose a more radical treatment strategy (a metaphorical hail mary). In such a case, it is hard to say whether the treatment strategy or the poorer starting positions of the patients is driving the poorer outcomes for a given strategy. More generally, if patients are not assigned at random, then we have no way of distinguishing the effect of the treatment from confounding variables. Even with appropriate randomization, one is left to wonder about how exactly the treatments were administered and enforced. With the two medications, this is not so problematic; the validity of the experiment simply hinges on subjects taking the instructed dose at the instructed times. In some sense, we do not expect much variability within these two groups

in how individuals deploy their treatment. But for the lifestyle treatment, what does it mean to "exercise and eat better"? Not only does this seem relative to the subject's starting point, but without explicit improvement criteria, the magnitude of improvement is also highly variable. In this way, I would guess that the treatment group is comprised of subgroups with varying degrees of commitment to lifestyle changes, and each of these groups may have meaningfully different success rates. To better understand lifestyle changes, I would therefore perform a follow-up experiment where I imposed more structural lifestyle changes as treatments. Another concern is the temporal scale. Staving off diabetes is a lifelong process, so what time scale exactly are we using to dictate success? It is not hard to imagine that certain medications or strategies may plateau, worsen, or possibly improve effectiveness with time.

Table 1: Summary Statistics of Treatments

Treatment	Subjects	Success	Failure	Success Rate
Lifestyle	234	125	109	53.41%
Metformin	232	112	120	48.27%
Rosiglitazone	233	143	90	61.37%

Figure 1: Bootstrapped KDE and ECDFs for Treatment Groups

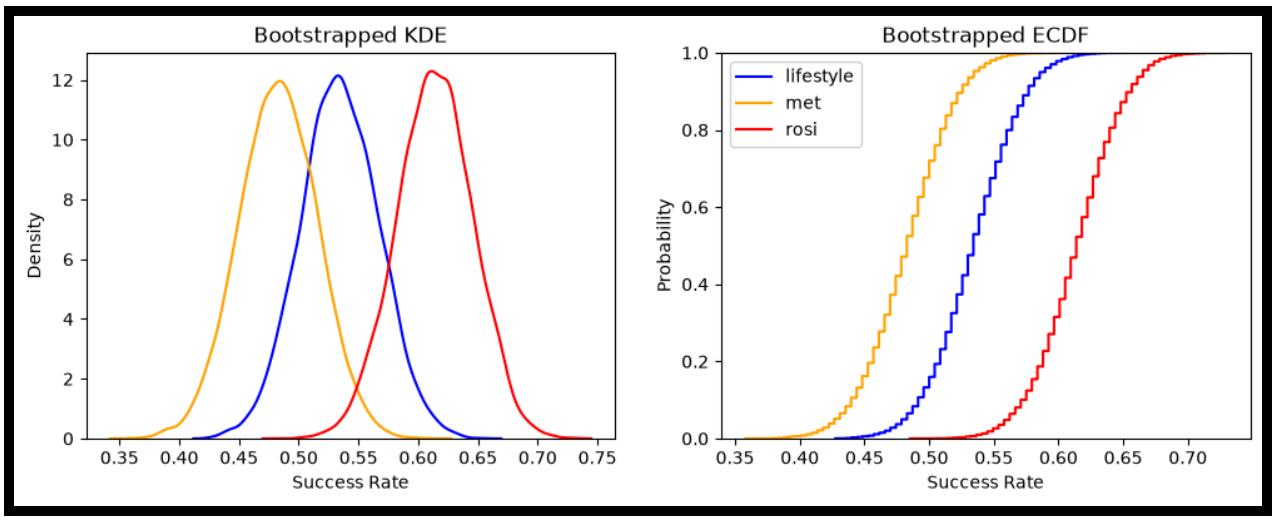


Figure 2: Bootstrapped KDE and ECDFs for ATE (Lifestyle - Metformin)

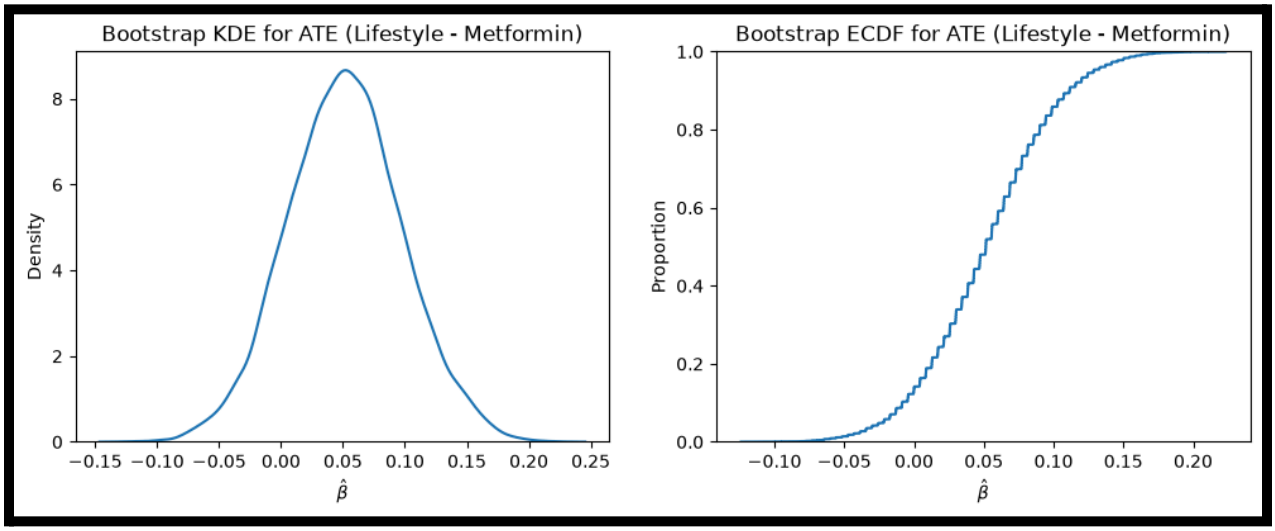


Figure 3: Bootstrapped KDE and ECDFs for ATE (Lifestyle - Rosiglitazone)

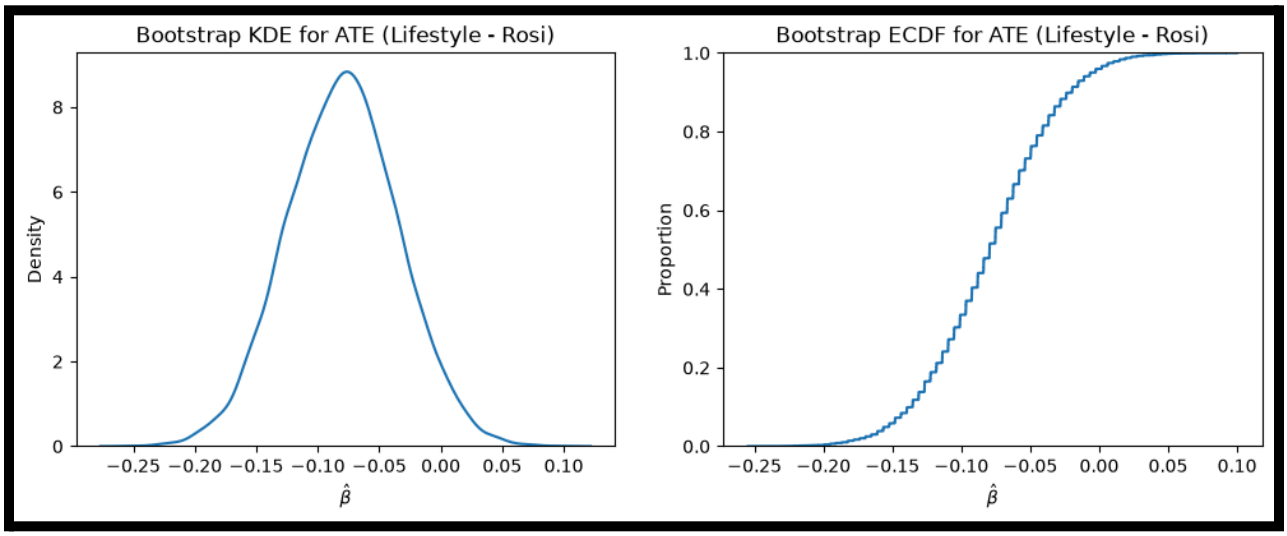


Figure 4: Bootstrapped KDE and ECDFs for ATE (Metformin - Rosiglitazone)

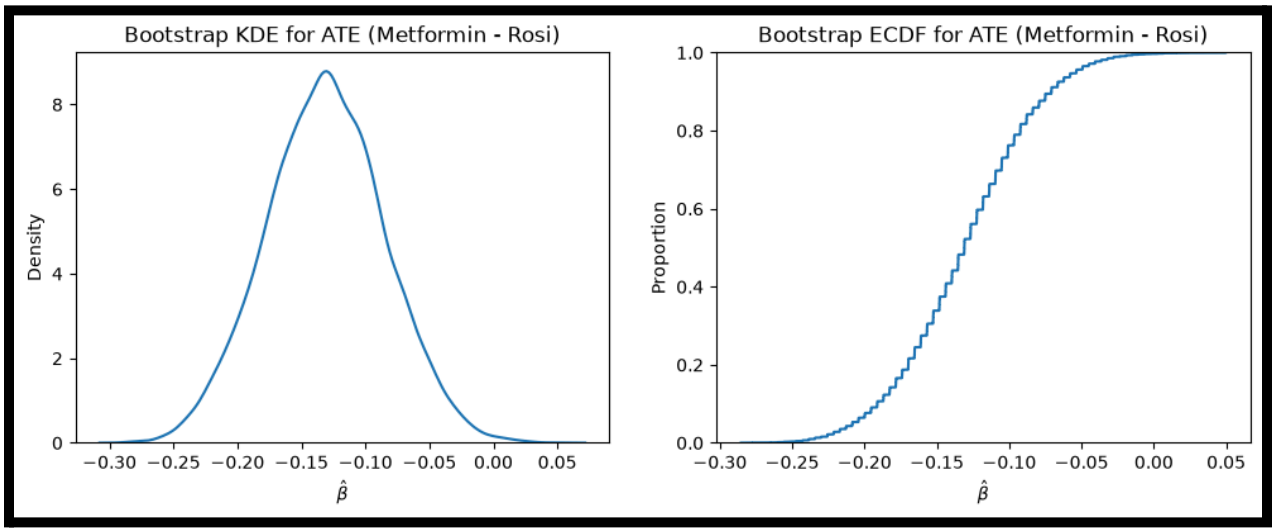


Table 2: Confidence Intervals for Average Treatment Effects

Average Treatment Effect	90% CI	95% CI	99% CI	Highest Significant Threshold
Lifestyle - Metformin	[-0.02169909, 0.12868406]	[-0.03864666, 0.14566847]	[-0.06469238, 0.17145631]	None
Lifestyle - Rosiglitazone	[-0.1524348, -0.00652856]	[-0.16549833, 0.01028988]	[-0.19524247, 0.0402775]	90%

Metformin - Rosiglitazone	[-0.208358, -0.05359257]	[-0.22127053, -0.04064304]	[-0.24291494, -0.01065552]	99%